

This information tells us what can and cannot be extracted via a linear probe, without having to train any probes. Furthermore, if we treat each  $\mathbf{g}_i$  as a *feature* (see Definition 1), then the linear regression coefficients tell us which directions in  $X_{i,l}$  these features are represented in, connecting back to Hypothesis 2.

Since  $X_{i,l}$  consists of modular addition problems with two inputs  $\alpha$  and  $\beta$ , we can visualize the errors as we iteratively construct  $\mathbf{g}_1, \dots, \mathbf{g}_k$  by making a heatmap with  $\alpha$  and  $\beta$  on the two axes, where the color shows what kind of error is made. More specifically, we take the top 3 PCA components of the error distribution and assign them to the colors red, green, and blue. We call the resulting heatmap a **residual RGB plot**. Errors that depend primarily on  $\alpha$ ,  $\beta$ , or  $\gamma$  show up as horizontal, vertical, or diagonal stripes on the residual RGB plot.

In Fig. 24, we perform EVR on the layer 17-29 hidden states of Mistral 7B on the Weekdays task; additional deconstructions are in Appendix K. We find that a circle in  $\gamma$  develops and grows in explanatory power; we plot the layer 25 residuals after explaining with one hot functions in  $\alpha$  and  $\beta$  (i.e.  $\mathbf{g}_1 = [\alpha = 0]$ ,  $\mathbf{g}_2 = [\beta = 1]$ ,  $\mathbf{g}_3 = [\alpha = 1], \dots$ ) in Fig. 23 to show this incredibly clear circle in  $\gamma$ . This suggests that the models may be generating  $\gamma$  by using a trigonometry based algorithm like the “clock” (Nanda et al., 2023a) or “pizza” (Zhong et al., 2024) algorithm in late MLP layers.

Table 4: Highest intervention effect attention heads from fine-grained attention head patching, as well as EVR results with one hot  $\alpha$ ,  $\beta$  and one hot  $\alpha$ ,  $\beta$ ,  $\gamma$ .

| (a) Mistral 7B, Weekdays. |    |                                        |                                         |                                                 | (b) Llama 3 8B, Weekdays. |    |                                        |                                         |                                                 |
|---------------------------|----|----------------------------------------|-----------------------------------------|-------------------------------------------------|---------------------------|----|----------------------------------------|-----------------------------------------|-------------------------------------------------|
| L                         | H  | Average<br>Inter-<br>vention<br>Effect | EVR $R^2$<br>One Hot<br>$\alpha, \beta$ | EVR $R^2$<br>One Hot<br>$\alpha, \beta, \gamma$ | L                         | H  | Average<br>Inter-<br>vention<br>Effect | EVR $R^2$<br>One Hot<br>$\alpha, \beta$ | EVR $R^2$<br>One Hot<br>$\alpha, \beta, \gamma$ |
| 28                        | 18 | 0.22                                   | 0.39                                    | 0.73                                            | 17                        | 0  | 0.18                                   | 0.98                                    | 0.99                                            |
| 18                        | 30 | 0.17                                   | 0.95                                    | 0.96                                            | 17                        | 1  | 0.08                                   | 0.98                                    | 0.98                                            |
| 15                        | 13 | 0.17                                   | 0.94                                    | 0.95                                            | 19                        | 10 | 0.08                                   | 0.95                                    | 0.96                                            |
| 22                        | 15 | 0.11                                   | 0.77                                    | 0.82                                            | 30                        | 17 | 0.07                                   | 0.85                                    | 0.90                                            |
| 16                        | 21 | 0.09                                   | 0.92                                    | 0.93                                            | 17                        | 3  | 0.07                                   | 0.93                                    | 0.95                                            |
| 28                        | 16 | 0.08                                   | 0.42                                    | 0.69                                            | 17                        | 27 | 0.06                                   | 1.00                                    | 1.00                                            |
| 15                        | 14 | 0.06                                   | 0.98                                    | 0.99                                            | 31                        | 22 | 0.05                                   | 0.37                                    | 0.78                                            |
| 30                        | 24 | 0.05                                   | 0.43                                    | 0.79                                            | 21                        | 9  | 0.04                                   | 0.73                                    | 0.78                                            |
| 21                        | 26 | 0.04                                   | 0.53                                    | 0.63                                            | 20                        | 28 | 0.04                                   | 1.00                                    | 1.00                                            |
| 14                        | 2  | 0.04                                   | 0.93                                    | 0.95                                            | 30                        | 16 | 0.04                                   | 0.73                                    | 0.85                                            |

| (c) Mistral 7B, Months. |    |                                        |                                         |                                                 | (d) Llama 3 8B, Months. |    |                                        |                                         |                                                 |
|-------------------------|----|----------------------------------------|-----------------------------------------|-------------------------------------------------|-------------------------|----|----------------------------------------|-----------------------------------------|-------------------------------------------------|
| L                       | H  | Average<br>Inter-<br>vention<br>Effect | EVR $R^2$<br>One Hot<br>$\alpha, \beta$ | EVR $R^2$<br>One Hot<br>$\alpha, \beta, \gamma$ | L                       | H  | Average<br>Inter-<br>vention<br>Effect | EVR $R^2$<br>One Hot<br>$\alpha, \beta$ | EVR $R^2$<br>One Hot<br>$\alpha, \beta, \gamma$ |
| 20                      | 28 | 0.15                                   | 0.76                                    | 0.76                                            | 15                      | 13 | 0.26                                   | 0.62                                    | 0.62                                            |
| 17                      | 0  | 0.10                                   | 0.77                                    | 0.77                                            | 16                      | 21 | 0.17                                   | 0.76                                    | 0.76                                            |
| 25                      | 14 | 0.08                                   | 0.19                                    | 0.61                                            | 18                      | 30 | 0.13                                   | 0.77                                    | 0.77                                            |
| 17                      | 1  | 0.07                                   | 0.80                                    | 0.82                                            | 28                      | 18 | 0.11                                   | 0.13                                    | 0.52                                            |
| 17                      | 3  | 0.06                                   | 0.71                                    | 0.71                                            | 28                      | 16 | 0.07                                   | 0.13                                    | 0.52                                            |
| 31                      | 22 | 0.06                                   | 0.12                                    | 0.67                                            | 21                      | 25 | 0.05                                   | 0.65                                    | 0.70                                            |
| 17                      | 27 | 0.05                                   | 0.58                                    | 0.58                                            | 15                      | 14 | 0.03                                   | 0.72                                    | 0.72                                            |
| 19                      | 4  | 0.05                                   | 0.40                                    | 0.66                                            | 17                      | 26 | 0.02                                   | 0.77                                    | 0.77                                            |
| 19                      | 10 | 0.04                                   | 0.62                                    | 0.62                                            | 31                      | 1  | 0.02                                   | 0.11                                    | 0.57                                            |
| 30                      | 26 | 0.04                                   | 0.51                                    | 0.62                                            | 21                      | 24 | 0.02                                   | 0.30                                    | 0.45                                            |

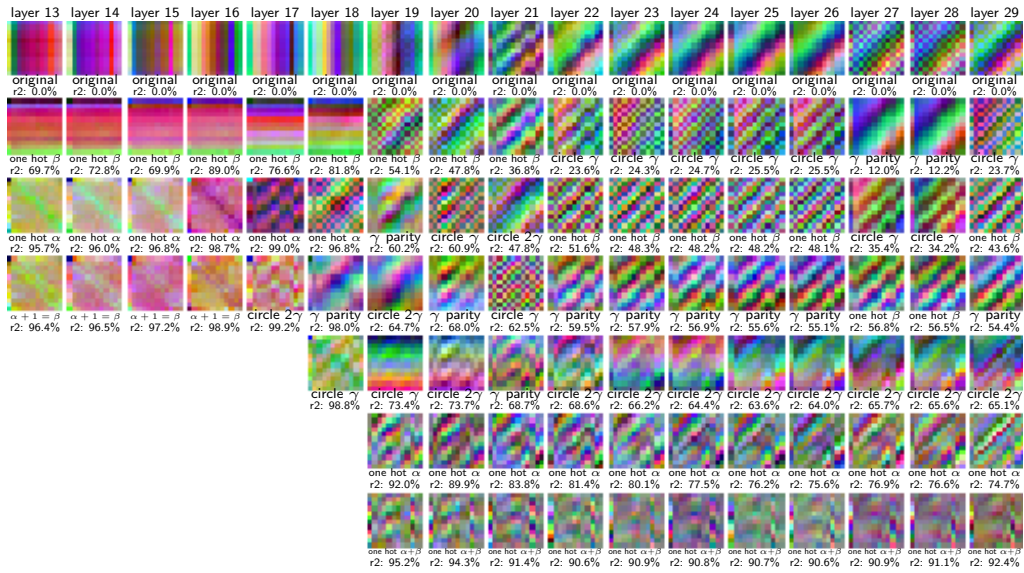


Figure 25: Iterative deconstruction of hidden state representations on the final token on Llama 3 8B, Months.